

State of the art report

- **IMPORTANT!** <https://zenodo.org/records/17885158> (REACTOME< but cool downloads)
- **IMPORTANT!** <https://reactome.org/download-data>
- <https://link.springer.com/article/10.1186/gb-2010-11-5-r53> ok WHAT??
- <https://www.liebertpub.com/doi/abs/10.1089/cmb.2023.0242>
 - <https://github.com/gkrieg/mmumin>
 - <https://github.com/annaritz/pathway-connectivity>
 - properly modeling the reaction networks that represent such pathways requires **directed hypergraphs** where each molecule or compound maps to a **vertex**, and each reaction maps to a **hyperedge** directed from its set of input reactants to its set of output products. Inferring the most likely series of reactions that produces a given set of targets from a given set of sources, where for each reaction its reactants are produced by prior reactions in the series, corresponds to finding a shortest hyperpath in a directed hypergraph, which is NP-complete.
 - We give the first exact algorithm for general shortest hyperpaths that can find provably optimal solutions for large, real-world, reaction networks. In particular, we derive a novel graph-theoretic characterization of hyperpaths, which we leverage in a new integer linear programming formulation of shortest hyperpaths that for the first time handles cycles, and develop a cutting-plane algorithm that can solve this integer linear program to optimality in practice.
 - We demonstrate that our cutting-plane algorithm quickly finds an optimal hyperpath—inferring the most likely pathway—with a median running time of under 10 seconds, and a maximum time of less than 30 minutes, even on instances with thousands of reactions. We also explore for the first time how well hyperpaths infer true pathways, and show that shortest hyperpaths accurately recover known pathways, typically with very high precision and recall.
- <https://ieeexplore.ieee.org/abstract/document/4605956> (2008)
 - Modeling systems biology from the point of view of discrete and hybrid systems
- **IMPORTANT!** <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1007384> (2019)
 - We examined the connectivity of Reactome signaling pathways using four different pathway representations. We find that Reactome is:
 - very **well** connected as a **graph**
 - **moderately well** connected as a **compound graph** or **bipartite graph**
 - **poorly** connected as a **hypergraph** (which captures many-to-many relationships in reaction networks)
 - We present a novel relaxation of hypergraph connectivity that iteratively increases connectivity from a node while preserving the hypergraph topology
- **IMPORTANT!** <https://academic.oup.com/nar/article/54/D1/D673/8326464> (Reactome 2026, published 2025... ok, figata)
 - https://watermark02.silverchair.com/gkaf1223.pdf?token=AQECAHi208BE49Ooan9kKhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAA1IwggNOBgkqhkiG9w0BBwagggM_MIIDOWIBADCCAzQGCSqGSIB3DQEHATAeBgIghkgBZQMEAS4wEQQMPluV_x8K4Vn0hzLAgEQgIIBDVv4BdY3wEpXPbU0eRegultn9XtbTZQt2xynKa-X-zafVmaSqyjlKo4hhu73lUI__rwoDdpQyLGDyxGZ3TKnFkqBjnJPJHEvvXLm3cW-NIpYabISXMgSbnYef6wpPCkUW8pZ5EX6FZoZ301Maq6FCrAL0_P53mzbrkzg9mku78CqkOiIWYB48FkWhLZA-saNXqpVZGmskzd2SspXlpgRnlbl_9iOyv6Fhtc6ipFvtCrzLEYiaMJjQGSHFZj7KBLfSJfDf1acsIgZ91Q70PaxC2-3ILWM1joB4c-VswgOkeT1QLklQslhsi92Zl-E1P0MppGGAfoMrGcF1quKbZabAoci5K-owK_INdCjJd37_nVWbCKgWsTreZxYvc7vFKcr9

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- SEE ALSO: <https://academic.oup.com/nar/article/52/D1/D672/7369850> (2024)
- SEE ALSO: <https://academic.oup.com/nar/article/50/D1/D687/6426058> (2022)
- <https://link.springer.com/article/10.1186/s13015-022-00217-9> (2022)
 - mi pare di averlo già visto?? Non è roba del problema di Linear Programming?
- https://link.springer.com/chapter/10.1007/978-3-642-20149-3_32 (2014)
 - reachability in a graph where edges have a certain “probability” to appear
- <https://www.biorxiv.org/content/10.1101/593913v2.abstract> (2019)
 - già visto ma su un altro sito (devo solo metterlo insieme)
- <https://link.springer.com/article/10.1186/1752-0509-8-30> (2014)
 - nah, non mi convince very much, potrebbe comunque essere utile
- https://link.springer.com/chapter/10.1007/978-3-031-65887-7_1 (2024)
 - cita i boolean network, con “dozzine” di cose... boh
- <https://ieeexplore.ieee.org/abstract/document/4402003> (2007)
 - SAME AS <https://scholarsarchive.byu.edu/facpub/225/>
 - (useless for me, at least from the abstract)

Making efficient queries

- <https://dl.acm.org/doi/abs/10.1145/1376616.1376677> (2008)
 - make efficient reachability queries (not for me)
- <https://dl.acm.org/doi/abs/10.1145/1871437.1871625> (2010)
- <https://dl.acm.org/doi/abs/10.1145/1989323.1989419> (2011)